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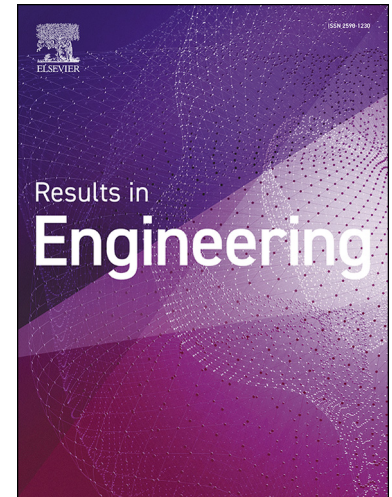
Extendable Switched-Capacitor Multilevel Inverter with DC Strings and Self-Balancing Capacitors

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Highlights

- A novel switched-capacitor multilevel inverter (SC-MLI) is proposed.
- The proposed SC-MLI can achieve high voltage gain with a reduced complexity.
- Self-balancing capacitors are used to enhance reliability and eliminate external control.
- Extendable topology enables scalable levels using direct current (DC) strings for flexible applications.
- High efficiency and voltage stability are validated through simulations and experiments.

Extendable Switched-Capacitor Multilevel Inverter with DC Strings and Self-Balancing Capacitors

Abstract

A multilevel inverter (MLI) is a critical component used to convert lower voltage levels to higher power loads in the field of renewable energy. By integrating switched-capacitor (SC) units with DC-AC converters, it can offer advantages over conventional two-stage conversion systems, including improved efficiency, optimal component count, and smaller passive element sizes for both inductors and capacitors. This paper introduces a novel extendable SC-MLI with the flexibility in cascading numbers of DC inputs and SC units to generate higher voltage levels and boost gain, making it an ideal topology for practical applications such as static synchronous compensators (STATCOMs). A STATCOM needs to generate different AC voltage amplitudes with low total harmonic distortion (THD) of 5%, as set by IEEE Standard 519-1992, across various modulation indexes (MI) to compensate for reactive power and enhance power system efficiency in modern networks. Moreover, the proposed topology requires lower-cost components, further enhancing its suitability for practical use. Then, the proposed topology is compared with the other existing extendable SC-MLIs in terms of the number of components, voltage stress, boost gain, cost, and conversion efficiency. Finally, a lab scaled experiment is conducted to verify the proposed topology with the PLECS-based simulation and theoretical analysis.

Keywords: Boost topology, dc-ac power conversion, dc strings, multilevel inverter, multi-source, switched-capacitor

1. Introduction

The topology of multilevel inverters (MLIs) has emerged as a viable solution for medium- to high-power conversion systems, including flexible AC power transmission systems, high-voltage direct current transmission systems, grid-connected solar PV systems, and high-power electric drives [1, 2].

6 MLIs are built using medium- to high-power semiconductors to generate a
 7 high-quality, stair-stepped sinusoidal AC output voltage waveform with re-
 8 duced total harmonic distortion and dv/dt stresses [3, 4]. This has made MLIs
 9 highly desirable, and researchers worldwide are working to improve their per-
 10 formance by addressing issues such as ripple of current and DC voltage bal-
 11 ancing and quality of the output signal enhancement [5, 6, 7]. Researchers
 12 are also focusing on achieving harmonic elimination through pre-calculated
 13 switching functions and developing new multilevel converter topologies and
 14 control strategies to comply with specific grid codes and utilize harmonic
 15 mitigation [8, 9, 10].

16 Conventional MLI topologies require linking with DC-to-DC boost con-
 17 verters in two-stage topologies to step up voltage sources, this setup making
 18 the overall system expensive and dissipating more of the source energy in
 19 high-voltage applications [11, 12, 13]. However, the existence of a switched-
 20 capacitor (SC) integrated multilevel inverter has mitigated these drawbacks
 21 [14]. Among the inverter topologies, the emergence of SC inverters demon-
 22 strates the benefit of being able to perform DC-DC boost functions and
 23 high-voltage-level DC-AC power conversions through a single-stage topol-
 24 ogy. It also offers better reliability due to the capability of self-balancing
 25 capacitor voltage, which is more applicable to high-power and high-boost
 26 gain conversion systems [14, 15, 16].

27 The wide applications of SC-based boost MLI (SC-MLI) topology have
 28 drawn interest from many researchers to further investigate and expose its
 29 structure with single or multiple DC voltage sources [17]. The fundamental
 30 operation of SC inverters is available in the literature [14, 15, 16, 17, 18],
 31 which can be further cascaded with the optimum selection of asymmetrical
 32 DC sources and DC capacitors to achieve high output gain and a number
 33 of voltage levels. For instance, the single-source SC-MLI, which generates
 34 49 voltage levels via 42 switches and 9 capacitors [19], is reconfigured in
 35 its switching states to produce 91 voltage levels with the same number of
 36 component counts [20] based on the nearest level control (NLC) modulation
 37 technique at the fundamental frequency. However, more active and passive
 38 components, as well as large total capacitor sizes ($42500\mu F$), are needed
 39 to achieve the aforementioned outcome, which in turn increased the system
 40 cost and losses (i.e., conduction, switching, and capacitor losses). Besides
 41 that, the employment of the complex voltage-balancing control algorithm
 42 to regulate the voltage of the capacitor resulted in a topology with a high
 43 computation burden and was difficult to replicate for the users.

Single-source SC-MLIs [21, 22, 23, 24, 25] post restrictions on producing filter-less or stair-stepped AC output voltage without employing an AC low pass filter. For instance, a single-source SC-MLI based on packed U-cell (PUC) proposed in [21] utilizes a switching-based optimized sliding-mode control (OSMC) to self-regulate/balance the capacitor voltage and drive the load in both grid-connected and stand-alone operation with a generated 7-level AC output voltage waveform. The topology is investigated under the dynamic mode, non-linear load, and parameters mismatch with a 20 kHz switching frequency. Although the number of passive (i.e., 1 unit of capacitor) and active components (i.e., 6 units of insulated-gate bipolar transistor (IGBT)) is significantly reduced with this configuration, the auxiliary circuit and voltage controller are still required to regulate the capacitor voltage. Worst of all, there is no boost function/conversion achieved by the topology. Moreover, the presented non-extendable topology requires the output LC filter to comply with a low total harmonics distortion (THD) of 5% set up by IEEE Standard 519-1992.

The emergence of multi-source SC inverters demonstrates the benefit of being able to generate a greater number of output voltage levels for lower THD (i.e., $\leq 5\%$). For instance, the SC-MLI proposed in [26] is capable of generating a 17-level output using dual asymmetrical DC inputs, ten switches, and two capacitors via the NLC method. Moreover, by employing cross-switched extension on parallel DC inputs with SC units, additional voltage levels such as 49-level and 137-level can be achieved. This module demonstrates the ability to generate high voltage levels while utilizing the minimum number of switches. However, as compared to the works in [19, 20], the proposed topology exhibits a limited boost gain of two. Additionally, the cross-switched extension results in a higher total switching voltage (TSV) experienced by the modules.

Hassan et al. [27] introduced a multi-source 21-level SC-MLI, constructed using K-type and H-bridge modules comprising fourteen switches dual H-bridge modules, two DC power supplies, and two capacitors. The presented SC inverter demonstrates the advantage of capacitor self-voltage balancing capabilities. However, the work still needs an H-bridge or a cascaded extension on the K-type module, to achieve additional voltage levels. This results in a higher TSV for the system. Additionally, the proposed topology exhibits a limited boost gain of 1.71. Furthermore, there are potential risks associated with unbalanced capacitor voltage due to asymmetrical output current in positive and negative half cycles. Another multi-source SC-MLI proposed

in [28] can produce 25 voltage levels and boost the input voltage by four using sixteen switches and two capacitors. The SC-MLI is capable of extending with different numbers of DC inputs and SC units to achieve desired voltage levels and boost gain. However, the required number of switches and associated voltage stresses increase significantly in proportion to the number of AC output voltage levels generated due to the use of dual bidirectional switches connected at the neutral point for multiple DC sources. For recently developed extendable SC-MLIs, these drawbacks are also highlighted in [29, 30, 31, 32]. In addition, the differences in nominal voltage between the separated/independent multiple DC voltage sources (i.e., voltage ratio) have to be taken into consideration when designing the asymmetrical SC-MLI. For instance, many SC-MLIs utilized the DC voltage sources with the voltage ratio of $(1, 3, 9, \dots)$, $(1, 5, 25, \dots)$ or higher [26, 27, 29]. Such ratios create unequal discharge rates among the multi-source and cause unequal power loss across the switch connected across each DC voltage source.

Additionally, recent research has investigated the development, implementation, and performance of switched-capacitor multilevel inverters in solar photovoltaic applications [33, 34] and advanced control strategies for DSTATCOM systems under non-ideal grid conditions [35], further demonstrating the versatility and significance of SC-MLI technologies in renewable energy integration and power quality improvement. All symmetrical and asymmetrical DC source configurations using the hybrid formed MLI (HMLI) topologies from the existing research works [26, 28, 27, 29, 30, 31, 32] are simply achieved by employing a higher number of independent battery packs or input sources with large dispersed voltage ratios. The addition of input sources prompts unequal power distribution and voltage balancing problems. In addition, the increase of complexity and cost may jeopardize the compactness factor and operation efficiency of the topology if taking into consideration the high number of passive components, active components, and DC-DC isolation in the circuitry (i.e., for the topology with limited boost gain). Moreover, the employment of the complex voltage-balancing control algorithm with the auxiliary circuit to regulate the capacitors voltage results in the topology with high computation and system cost. Furthermore, numerous non-extendable topologies generate AC output through low-pass filters or cascaded extensions, making the prototype expensive for high-voltage applications such as STATCOM. STATCOM needs to generate different AC voltage amplitudes or voltage levels across various *MI* to compensate for reactive power, thereby maximizing power system efficiency in

modern networks while complying with IEEE Standard 519-1992 [36].

This paper proposed an extendable configuration of SC-MLI. Therefore, the need for asymmetrical DC sourcing and other aforesaid issues for HMLI to obtain high-level AC output voltage as well as high voltage gain can be solved. The main features of the proposed topology are as follows:

- The topology can be extended with basic DC sources and SC units for higher voltage levels and boosted gains. The use of DC strings formed by multi-source with the same voltage amplitude connected in series facilitates the creation of high voltage levels and helps alleviate imbalanced discharge rates among the DC sources, which could cause imbalanced power losses across the switches in the conducting path. Moreover, it eliminates the need for power devices connected between the DC sources, thereby reducing capacitor power losses from the charging current across the switches. Additionally, the DC strings reduce the number of capacitors required for multilevel output.
- The SC unit, comprising upper and lower leg capacitors, alternately charges and discharges, thereby minimizing discharging cycles and extending charging cycles. This effectively reduces the high supplying current required for charging from the DC source. Furthermore, the series-parallel combination of switched capacitors is efficiently utilized to reduce the count of DC sources and boost the voltage without requiring an auxiliary circuit. This balanced approach between multi-source and switched capacitor cells aims to reduce component count, boost capability, and overall system size and cost.
- A simple control scheme can be used to drive the proposed topology, and hence, it has less computation burden in the microcontroller. For instance, it requires only the Nearest Level Control (NLC) method to self-regulate the DC voltage across each DC capacitor.
- The proposed extendable SC-MLI requires components with lower costs for generating high-power and voltage-level output. This results in a prototype that is both inexpensive and compact compared to non-extendable topologies while pursuing efficient operation, making it suitable for a wide range of high-power applications. For instance, the proposed topology can generate 25 levels to ensure compliance with IEEE Standard 519-1992, achieving a targeted THD of $\leq 5\%$. Additionally,

it provides four times the boost gain without the need for a conventional DC-DC boost converter by utilizing only three series-connected symmetrical DC sources, twelve switches, four diodes, and four capacitors. With a greater number of DC sources and SC units, the proposed topology can generate more voltage levels and boost gain to meet the requirements of high-power applications such as STATCOM.

Table 1: Parameters for the proposed SC-MLI in DC strings configuration.

Parameters	DC Strings Configuration
Output levels	$N_L = 2^{n+1}m + 1$
DC sources	$N_{DC} = m$
Switches	$N_{sw} = 2m + 2n + 2$
Diode	$N_d = 2n$
Capacitor	$N_c = 2n$
Components	$N_{comp} = 3m + 6n + 2$
Gate driver	$N_{gd} = m + n + 2$
Input voltage	$V_{a,j} = V_{a,1}$ for $j = 2, 3, \dots, m$
Total input voltage	$V_{in,total} = V_{a,1} + V_{a,2} + V_{a,3} + \dots + V_{a,m}$
Voltage boost gain	$V_B = 2^n$
Capacitor voltage	$V_{C,n} = 2^{j-1} \sum_{i=1}^m V_i$ for $j = 1, 2, 3, \dots, n$
AC output peak voltage	$V_{o,peak} = 2^n m V_{in}$
Total standing voltage	For switch:
	$TSV_{p.u.} = (m + 1) + 2 \times 2^{j-1} \sum_{i=1}^m V_i + 2 \times 2^n$
	for $j = 1, 2, 3, \dots, n$
	For diode:
	$TSV_{p.u.} = 2 \times 2^{j-1} \sum_{i=1}^m V_i$
	for $j = 1, 2, 3, \dots, n$

2. Extendable SC-MLI Topology

The existing SC-MLIs [21, 22, 23, 26, 27, 29] produce AC output voltage with fewer than 17 levels, resulting in a THD greater than 5%. In order to

attain higher output voltage levels, it is required to extend its modules with cascaded, cross-switched, and other configurations. It can ensure the compliance of IEEE Standard 519-1992 with the targeted $\text{THD} \leq 5\%$. However, the extended modules require a much higher DC value as their second source setup as compared to the first DC source. The ratios of $(1, 3, 9, \dots)$, $(1, 5, 25, \dots)$, $(1, 7, 49, \dots)$ in DC voltage cause the imbalance of discharge rates among the DC sources and subsequently, produce imbalanced power losses across the switches in the conduction path of DC sources.

A novel extendable SC-MLI with DC strings configuration and self-balancing switched-capacitor units is illustrated in Fig. 1. The DC strings and SC unit are composed of the number of DC sources (m) and a pair of capacitors (n). Each DC source V_m is formed by bidirectional switches $S_{a,m-1}$ whereas the switched-capacitor units ($C_{d,n}, C_{u,n}$) are formed by the pair of switches ($S_{c,n}, S'_{c,n}$) and diodes ($D_{c,n}, D'_{c,n}$). The half-bridge on the left-hand side ($S_{L,1}, S_{L,2}$) and the right-hand side ($S_{R,1}, S'_{R,1}$) are used for DC to AC conversion.

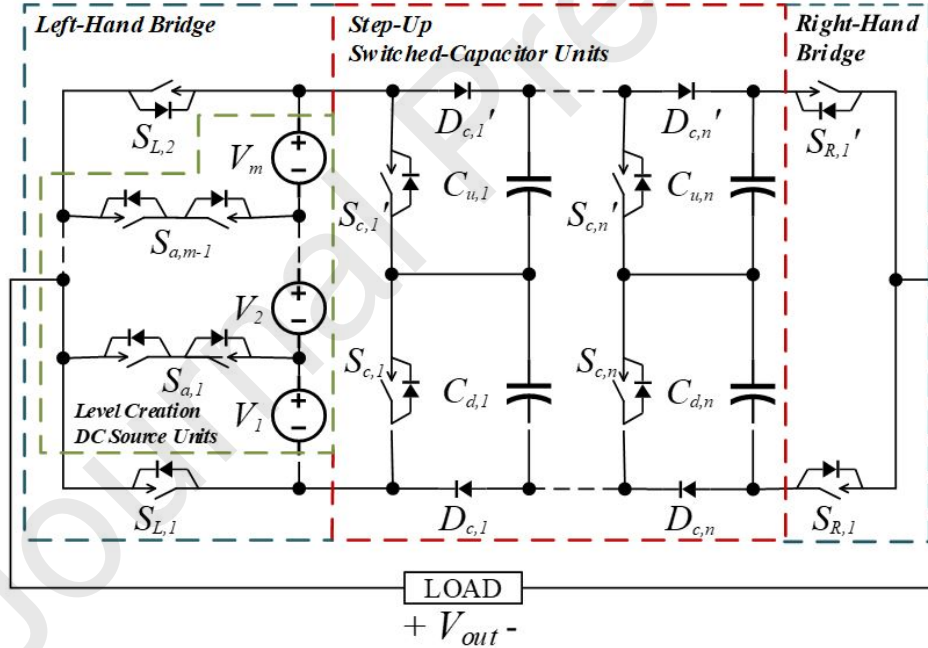


Figure 1: Extendable structure of the proposed DC strings based SC-MLI.

To prevent the short circuit of DC sources and capacitors, the pair of switches ($S_{c,n}, S'_{c,n}$) and ($S_{R,1}, S'_{R,1}$) and diode ($D_{c,n}, D'_{c,n}$) must not be

182 turned ON at the same time. It is also noteworthy that DC strings can be
 183 represented as the series-connected DC sources or battery packs with the
 184 same input voltage, neglecting the power imbalance among the isolated DC
 185 sources when applied to generate multilevel output. It also requires no power
 186 devices connected in between the DC sources and thus neglected the capacitor
 187 power losses from the charging current across the switches. Additionally,
 188 step-up capacitors from the DC strings-based SC-MLI are charged and self-
 189 balanced in a binary-asymmetric pattern to facilitate higher output voltage
 190 creation and boost gain. Table 1 lists down the components and parameters
 191 involved in achieving these outcomes.

192 3. Operation Principles

193 To achieve a 25-level output and boost up the input sources by four, the
 194 proposed SC-MLI requires configuration with only three series-connected DC
 195 sources with the same voltage magnitude to form DC strings and two double-
 196 mode SC units. The schematic of the proposed topology is shown in Fig. 1.

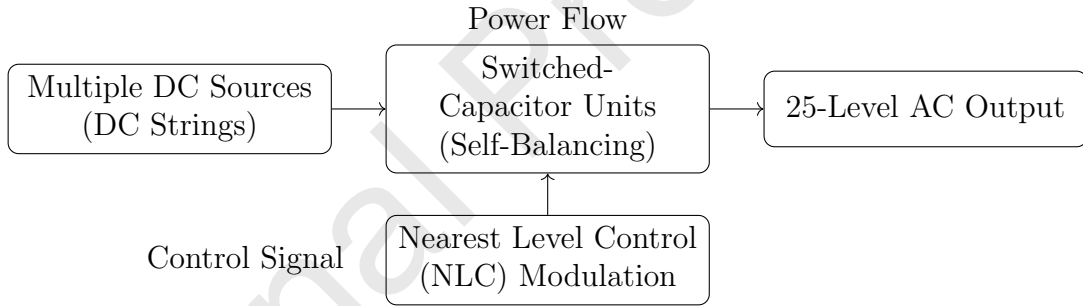


Figure 2: Block diagram of the proposed SC-MLI system with NLC modulation generating 25-level AC output.

197 Figure 2 illustrates the overall architecture of the proposed switched-
 198 capacitor multilevel inverter (SC-MLI) system. The design employs multiple
 199 series-connected DC sources, referred to as DC strings, which supply the
 200 input voltage to the SC-MLI power stage. This power stage integrates the
 201 series-parallel connected switched-capacitor units and the output bridge re-
 202 sponsible for generating the multilevel AC waveform. A NLC modulation
 203 strategy is implemented to determine the appropriate switching states of all
 204 devices, thereby synthesizing the desired 25-level staircase output voltage.

205 The detailed switching states of the proposed SC-MLI are shown in Fig.
 206 3. It indicates the ON and OFF states of the switches and diodes, as well
 207 as capacitor charging or discharging states. Notably, the NLC modulation
 208 technique is utilized to operate the proposed 25-level DC strings-based SC-
 209 MLI. It compares the reference sinusoidal waveform with precomputed volt-
 210 age levels to select the closest output level at each switching instant. This
 211 method is particularly well-suited for multilevel inverters with a high num-
 212 ber of voltage steps, as it requires significantly less computation compared
 213 to modulation strategies such as Selective Harmonic Elimination (SHE) or
 214 Phase-Shifted PWM. The use of NLC reduces the burden on the microcon-
 215 troller, enabling faster response times and efficient signal generation, which
 216 is critical when operating with higher voltage levels especially 17 levels or
 217 above. Compared to methods like SHE and PWM, NLC offers clear advan-
 218 tages in terms of computational simplicity and speed, which translates into
 219 lower processing costs and more straightforward implementation on low-cost
 220 digital controllers. However, a practical limitation of NLC is that at lower
 221 voltage levels below 17 levels, the output waveform deviates more visibly
 222 from a pure sine wave, leading to higher harmonic distortion. Fig. 3 also
 223 showing the switching logic sequences and capacitor self-voltage balancing in
 224 one cycle of output voltage. Here, $\Delta V_{cd,1}$, $\Delta V_{cu,1}$, $\Delta V_{cd,2}$, and $\Delta V_{cu,2}$ indicate
 225 the voltage ripple of capacitors $C_{d,1}$, $C_{u,1}$, $C_{d,2}$, and $C_{u,2}$, respectively. Fig. 4
 226 shows the detailed switching states of the proposed 25-level DC strings based
 227 on SC-MLI for the 1:1:1 V_{in} DC configuration with positive voltage states.

228 The proposed topology can be extended by adding more voltage sources
 229 and capacitors to achieve higher levels and boost gain. In the primary stage,
 230 DC strings are utilized to generate multilevel output. Then, in the sec-
 231 ondary stage, the capacitors $C_{d,1}/C_{u,1}$ in the first SC unit are pre-charged
 232 alternately during positive/negative voltage states using the entire DC in-
 233 put voltage ($V_{in,total} = V_1 + V_2 + V_3$) through a series-parallel combination.
 234 This process increases the voltage level and boost gain ($V_B=2$). Once the
 235 capacitors in the first SC unit are charged and equivalent to the total DC
 236 input voltage ($V_{c,d1} = V_{c,u1} = V_{in,total}$), the capacitors $C_{d,2}/C_{u,2}$ in the sec-
 237 ond SC unit are further charged by the series-connected DC inputs and the
 238 first SC unit ($V_{c,d2} = V_{in,total} + V_{c,u1}$ or $V_{c,u2} = V_{in,total} + V_{c,d1}$) alternately
 239 during positive/negative voltage states. These repeating/interchangeable
 240 charging and discharging steps create a binary asymmetrical pattern, en-
 241 suring the capacitors self-balance their voltage in every cycle. Finally, the
 242 DC strings and SC units are connected in series to achieve the highest volt-

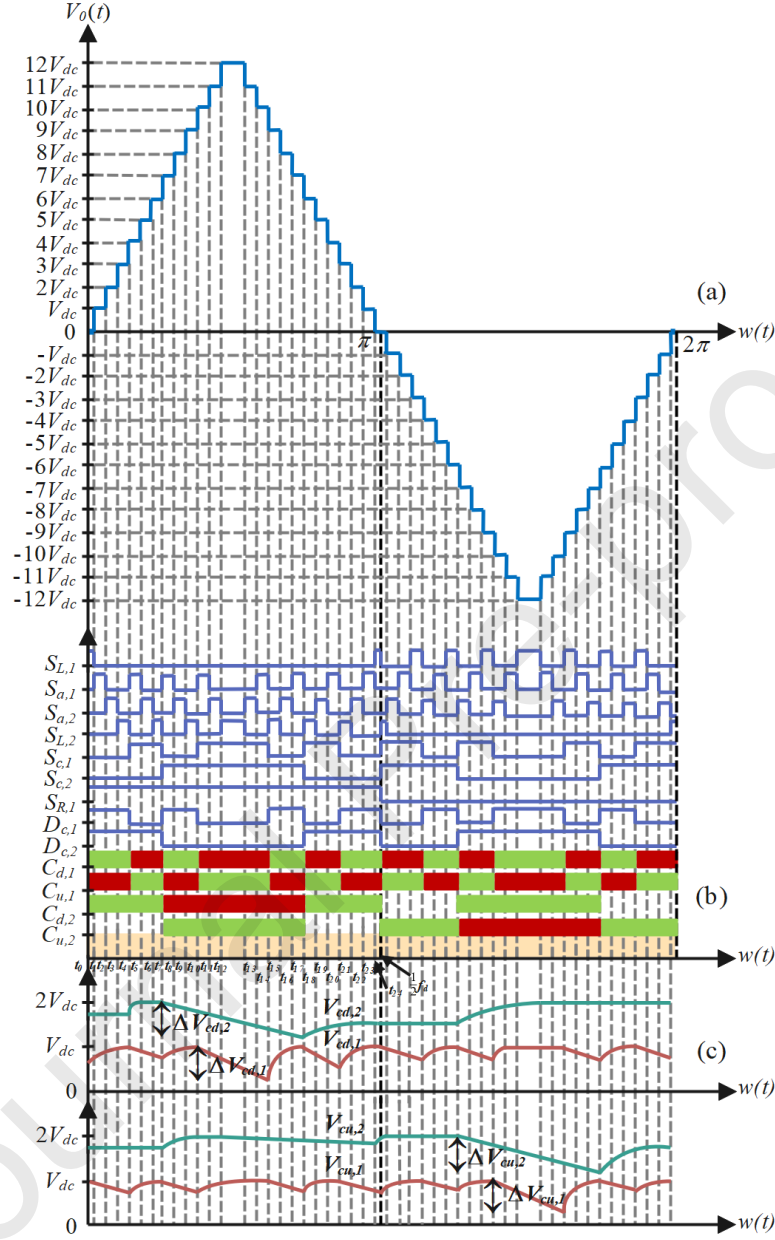


Figure 3: (a) NLC modulation for proposed 25-level SC-MLI showing (b) switching logic sequences and (c) capacitor self-voltage balancing in one cycle of output voltage. Note: In (b): green = charging, red = discharging.

age level ($V_{o,peak} = V_{in,total} + V_{c,d1} + V_{c,d2}$ for positive output voltage level and $V_{o,peak} = V_{in,total} + V_{c,u1} + V_{c,u2}$ for negative output voltage level) with gain ($V_o = 4V_{in,total}$ or $V_B = 4$). Lastly, the multilevel DC output voltage is inverted to AC through half-bridges on the left-hand and right-hand sides of the topology.

Table 2: Comparison of extendable multisource SC-MLIs with proposed topology for particular AC output voltage level from 17-level to 25-level

Ref	N_L	N_{DC}	N_{sw}	N_d	N_C	N_{comp}	N_{gd}	V_B	$TSV_{p.u.}$		CF			V_{in} ratio	Extended with
									Switch	Diode	$\alpha=0.5$	$\alpha=1.0$	$\alpha=1.5$		
[26]	17	2	10	2	2	16	5	2	5	0.5	22.38	23.75	25.13	1:3	DC input and cross-switches
[27]	21	2	14	0	2	18	11	1.71	N/A	N/A	N/A	N/A	N/A	1:6	H-bridge units and cascade
[28]	25	2	16	0	2	20	10	4	6.75	0	30.84	31.69	32.53	1:2	DC input and SC units
[29]	25	2	13	0	2	17	12	2	10.5	0	31.63	34.25	36.88	1:4	DC input and SC units
[30]	25	2	16	0	2	20	10	4	6.8	0	30.85	31.70	32.40	1:2	Cross-switches DC input and SC units
[31]	19	3	14	3	3	23	8	1.57	5.82	0.64	33.06	35.11	37.17	1:2:4	DC input and SC units
[32]	21	2	16	1	2	19	11	3.33	6.5	0.2	31	32.01	33.02	1:2	DC input and SC units
Ours	25	3	12	4	4	23	7	4	4.5	1.5	30.75	31.50	32.25	1:1:1	DC strings and SC cells

4. Comparative Study

A comparative study is performed between proposed topologies and existing developed SC-MLIs [26, 28, 27, 29, 30, 31, 32] in terms of their design parameters. The proposed topology is evaluated for its number of components and gate drivers (N_{gd}), boost gain (G_b), total standing voltage per unit ($TSV_{p.u.}$), cost function (CF), and the DC input voltage deviation ratios. The extendability features are benchmarked against the existing extendable SC-MLIs for AC output voltage levels ranging from 17 to 25 levels. To determine the design superiority of the SC-MLIs, the definition of CF is given by

$$CF = N_{DC} + N_{sw} + N_d + N_C + N_{gd} + \frac{\alpha \times TSV_{p.u.}}{V_B}, \quad (1)$$

where α the weight coefficient which is set to be 0.5, 1.0, and 1.5. A smaller weight coefficient α places more emphasis on component counts, while a bigger weight coefficient α prioritizes TSV. As shown in Table 2, our topology exhibits lower CF compared to [28, 27, 29, 30, 31, 32]. While [26] has the least CF, its boost gain of two and high voltage deviation ratio (i.e., $1:3V_{in}$) pose limitations. Similar to the works of Hassan et al. [27], Bana et al. [29], and Singh et al. [31], its restricted boost gain is less than 2, and a high voltage deviation ratio of more than $1:3V_{in}$ limits its practical applicability. All the topologies, including our posed one, shown in Table 2 have a high

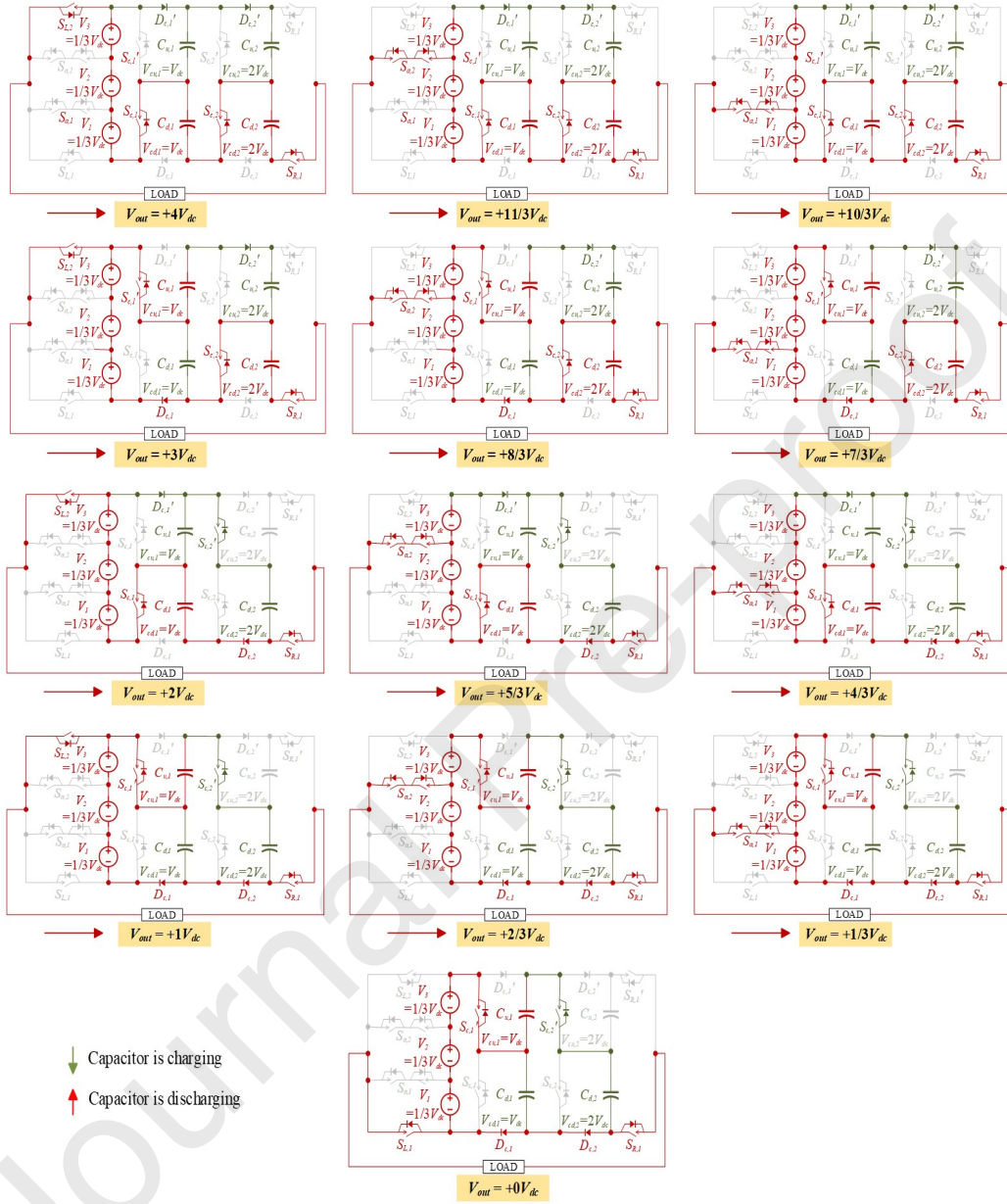


Figure 4: Detailed switching states of the proposed 25-level DC strings based on SC-MLI for 1:1:1 V_{in} DC configuration with positive voltage states.

267 boost gain of 4, with minimal voltage deviation, making them superior for
 268 use in renewable industrial applications. Additionally, our proposed SC-MLI

is designed using DC strings without the need for any power devices between DC sources, resulting in higher efficiency as the power sharing and capacitor charging current among the power devices is minimized.

Table 3: Cost comparison of SC-MLIs with proposed topology in DC strings configuration for particular AC output voltage levels from 17-Level to 25-Level

Components	Part Number	Rating	Unit Price* (\$)	[26] <i>G</i> -2	[28] <i>G</i> -4	[29] <i>G</i> -2	[30] <i>G</i> -4	[31] <i>G</i> -1.57	[32] <i>G</i> -3.33	Ours <i>G</i> -4
MOSFET	IRF540NPBF (TO-220)	100V, 33A	1.33	4	9	4	9	8	9	8
	IRFB4615PBF (TO-220)	150V, 33A	1.93	0	0	2	0	0	0	0
	IRFP250MPBF (TO-220)	200V, 30A	2.42	4	3	3	3	2	3	2
	IRFP360LCPBF(TO-247)	400V, 23A	5.66	2	4	4	4	4	4	2
Diode	STF12100 (ITO)	100V, 15A	0.28	1	0	0	0	2	1	2
	SDURF1520 (ITO)	200V, 15A	0.28	1	0	0	0	1	0	2
Capacitor	KY50VB102M12X40LL	50V, 1000 μ F	0.29	0	0	1	0	0	0	0
	380LQ102M100H022	100V, 1000 μ F	1.68	1	1	0	1	2	1	2
	380LX102M200J452	200V, 1000 μ F	2.55	1	1	1	1	1	1	2
Driver	HCPL-3120-060E	625V _{max}	3.32	6	10	12	10	8	11	7
Heatsink	WA-T220-101E (TO-220)	-	2.26	8	12	9	12	10	12	10
	WA-T247-101E (TO-247)	-	2.59	2	4	4	4	4	4	2
Total Cost (\$)	With Heatsink			73.80	116.00	111.00	116.00	103.67	119.39	86.77
	Without Heatsink			50.50	78.40	80.70	78.40	70.71	81.91	58.99

*Courtesy: www.digikey.com. Prices are subject to change based on market conditions. *G* is defined as voltage gain.

The cost assessment is an important evaluation to understand the total design expenditure based on the current commercialization market. In Table 3, the components are selected from the Digikey site to perform a fair comparison between the proposed topology and the recently developed extendable SC-MLIs for specific AC output voltage levels ranging from 17-Level to 25-Level. The proposed topology can be constructed at a lower price of \$58.99 as compared to [28, 27, 29, 30, 31, 32]. On the other hand, Barzegarkhoo and Kojabadi [26] proposed the extendable SC-MLIs with the lowest cost; however, the design has only two boost gains, making it less attractive for practical implementation. It is noteworthy that the proposed SC-MLI employs a DC strings configuration without the need for additional power devices between the DC sources. Consequently, it is expected to achieve the highest efficiency, given the minimized power-sharing and capacitor charging current among the power devices.

286 5. Simulation Validation

287 The proposed DC strings-based SC-MLI (25-level) is simulated using
 288 PLECS, with specifications outlined in Table 4, where three DC sources
 289 with $V_1 = V_2 = V_3 = 16.6\text{V}$ and two double-mode SC units with $C_{d,1} =$
 290 $C_{u,1} = 3300\text{ }\mu\text{F}$, $C_{d,2} = C_{u,2} = 2200\text{ }\mu\text{F}$ are utilized to provide 25-level out-
 291 put and boost up input voltage by four. Capacitance values chosen are
 292 also considered to minimize capacitor voltage ripple ($\leq 10\%$) due to dis-
 293 charging cycle. The fundamental frequency is chosen as 50Hz . The per-
 294 formance of the proposed SC-MLI is evaluated under two loading conditions,
 295 i.e., $R = 80\Omega$ and $R = 80\Omega$ and $L = 167.5\text{mH}$ utilizing real data from the de-
 296 vice manufacturer for switches and diodes to ensure practical outcomes. The
 297 switches and diodes used in this simulation are C3M0025065K (650V/97A)
 298 and C3D16060D (600V/22A), respectively. The turn-on-resistance of the
 299 switch and diode are $R_{on} = 25\text{m}\Omega$ and $R_d = 0.8265\text{m}\Omega$, respectively. The
 300 forward voltage of diode is $V_F = 1.6\text{V}$. The internal resistance of capacitors
 301 are $R_{c,d1} = R_{c,u1} = 23.5\text{m}\Omega$, $R_{c,d2} = R_{c,u2} = 30.0\text{m}\Omega$ for capacitors $C_{d,1}$, $C_{u,1}$,
 302 $C_{d,2}$ and $C_{u,2}$, respectively.

303 Fig. 5 depicts the waveforms of AC output voltage (V_{out}) and current
 304 (I_{out}), along with capacitor voltage ($V_{c,d1}$, $V_{c,u1}$, $V_{c,d2}$, and $V_{c,u2}$) for capacitors

Table 4: Specification of the Proposed Topology (25-level) in Simulation.

Description	Value/Parameters
Input DC source	$V_1 = V_2 = V_3 = 16.6\text{V}$
Boost gain	$V_b = 4$
Fundamental frequency	$f = 50\text{ Hz}$
Loads	R load: 80Ω $R - L$ load: 80Ω and 167.5mH
Capacitor	$C_{d,1} = C_{u,1} = 3300\text{ }\mu\text{F}$, $C_{d,2} = C_{u,2} = 2200\text{ }\mu\text{F}$
Internal resistance of capacitors	$R_{c,d1} = R_{c,u1} = 23.5\text{m}\Omega$, $R_{c,d2} = R_{c,u2} = 30.0\text{m}\Omega$
On-resistance of switch	$R_{on} = 25\text{m}\Omega$
Forward voltage of diode	$V_F = 1.6\text{V}$
On-resistance of diode	$R_d = 0.8265\text{m}\Omega$

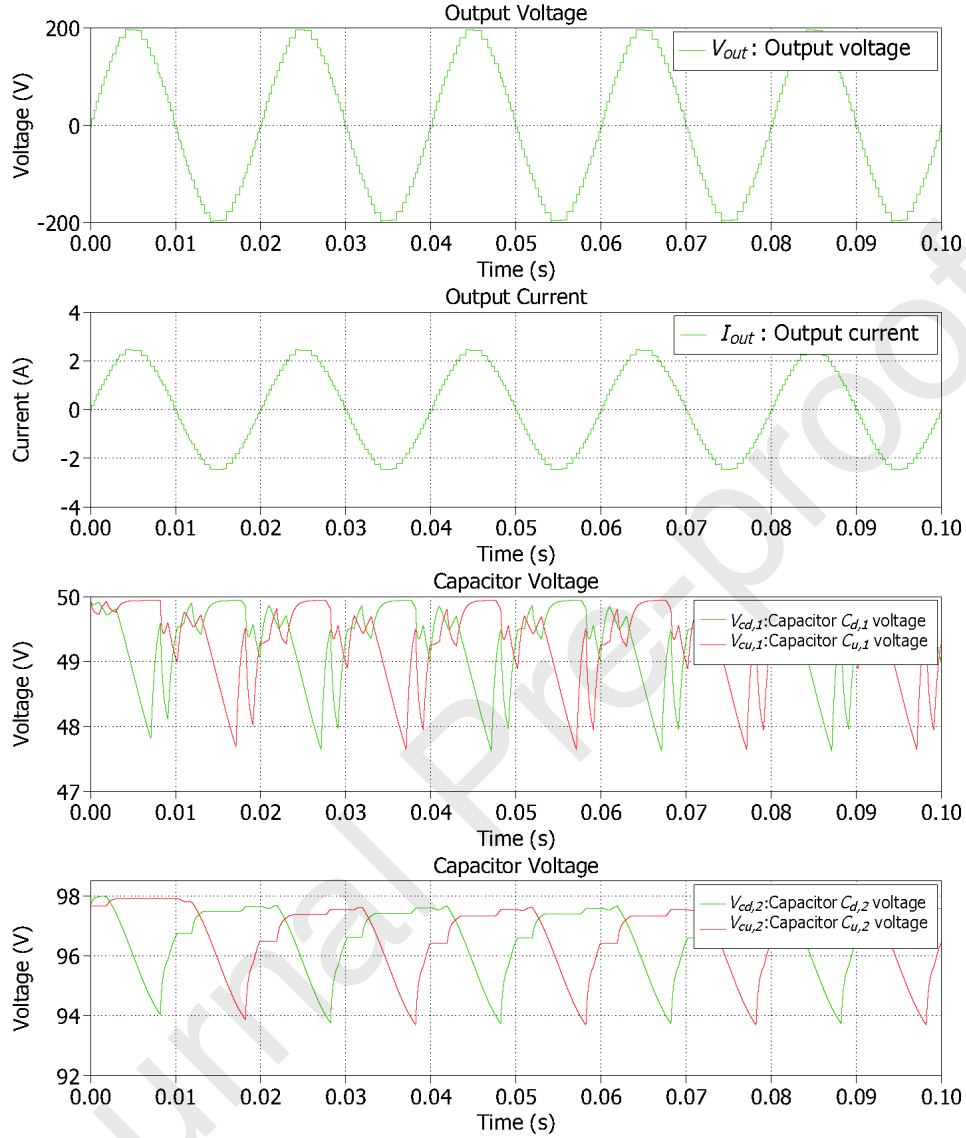


Figure 5: Simulation results of output voltage V_{out} , current I_{out} , and capacitor voltage $V_{c,d1}$, $V_{c,u1}$, $V_{c,d2}$, and $V_{c,u2}$ of the proposed SC-MLI in 25-level DC strings configuration with resistive load $R = 80\Omega$.

305 $C_{d,1}$, $C_{u,1}$, $C_{d,2}$ and $C_{u,2}$ under pure resistive load ($R = 80\Omega$). These figures
 306 demonstrate the effective conversion of DC inputs into 25-level AC output
 307 waveforms (V_{out} and I_{out}) by the proposed SC-MLI, resulting in a boost of
 308 the total input voltage ($V_{in,total}$) from 50V to a peak AC output voltage

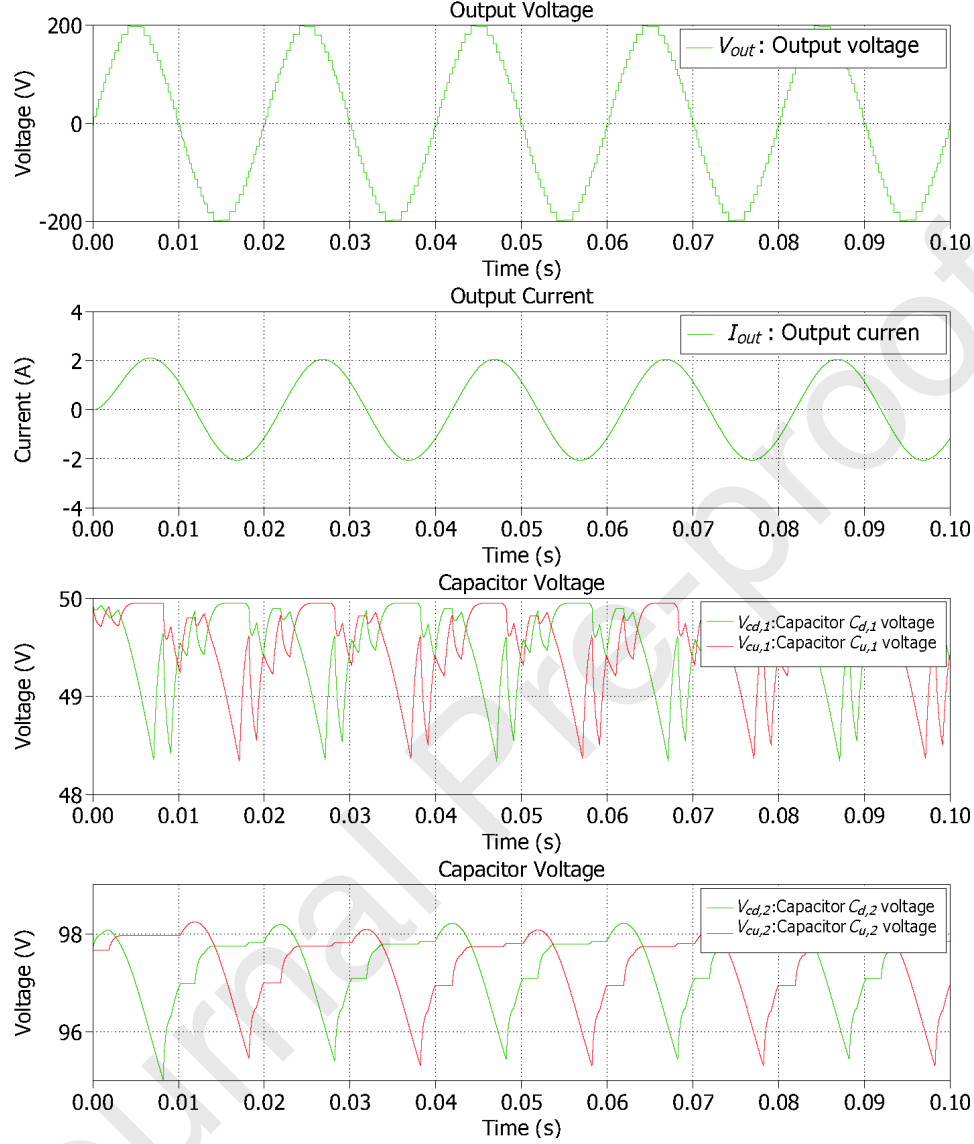


Figure 6: Simulation results of output voltage V_{out} , current I_{out} , and capacitor voltage $V_{c,d1}$, $V_{c,u1}$, $V_{c,d2}$, and $V_{c,u2}$ of the proposed SC-MLI in 25-level DC strings configuration with resistive-inductive load $R - L = 80\Omega + 167.5\text{mH}$.

309 ($V_{o,peak}$) of approximately 200V. The resultant output voltage and current
 310 measure $140.04V_{rms}$ and $1.72I_{rms}$, respectively. Additionally, the THD values
 311 for both the 25-level output voltage and current are found to be 3.75% and

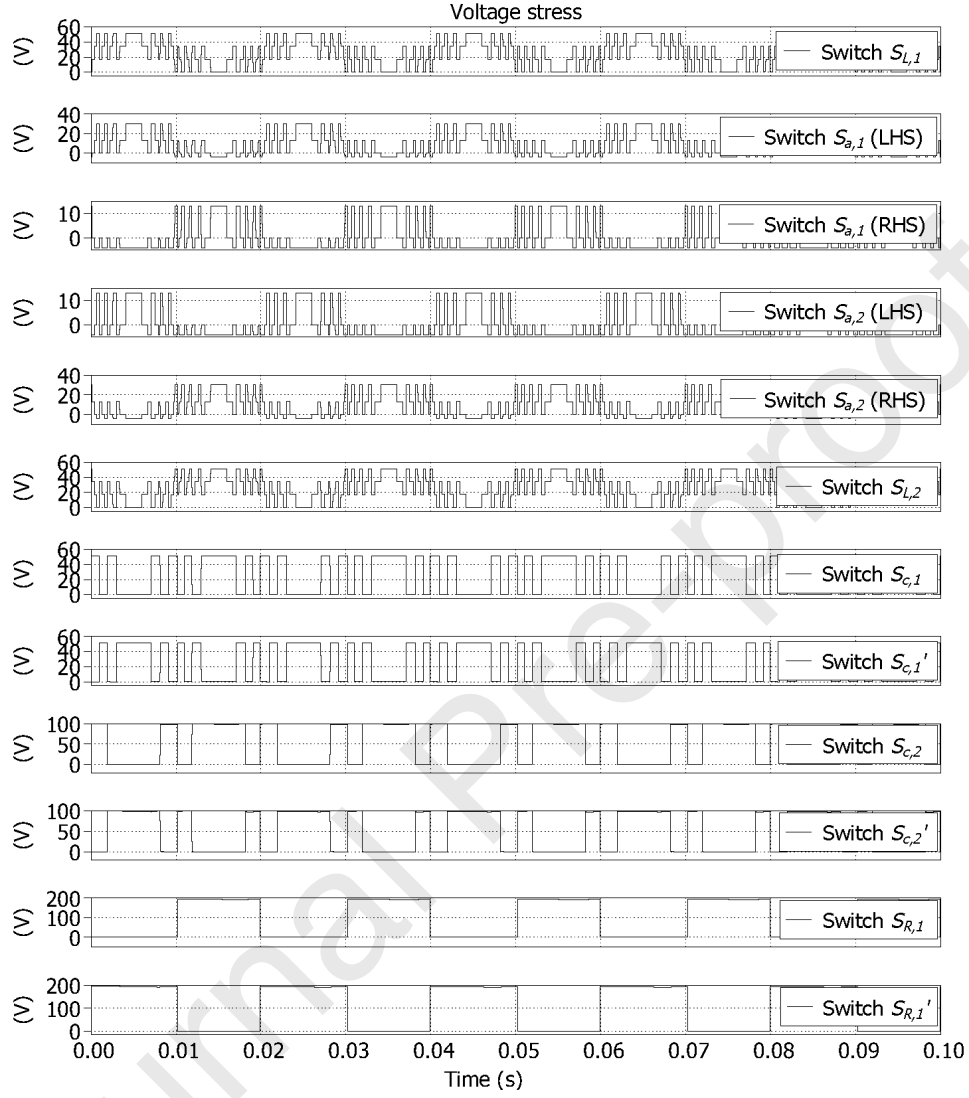


Figure 7: Simulation results of the voltage stress for the proposed SC-MLI in 25-level DC strings configuration.

312 3.59%, respectively, meeting IEEE Standard 519-1992 ($\text{THD} \leq 5\%$). The
 313 capacitors able to self-balancing its voltage around their respective values
 314 ($V_{c,d1} = V_{c,u1} = 50V$ and $V_{c,d2} = V_{c,u2} = 50V$) without the need for auxiliary
 315 circuits or control systems.

316 Similarly, Fig. 6 shows the output of the proposed SC-MLI generating

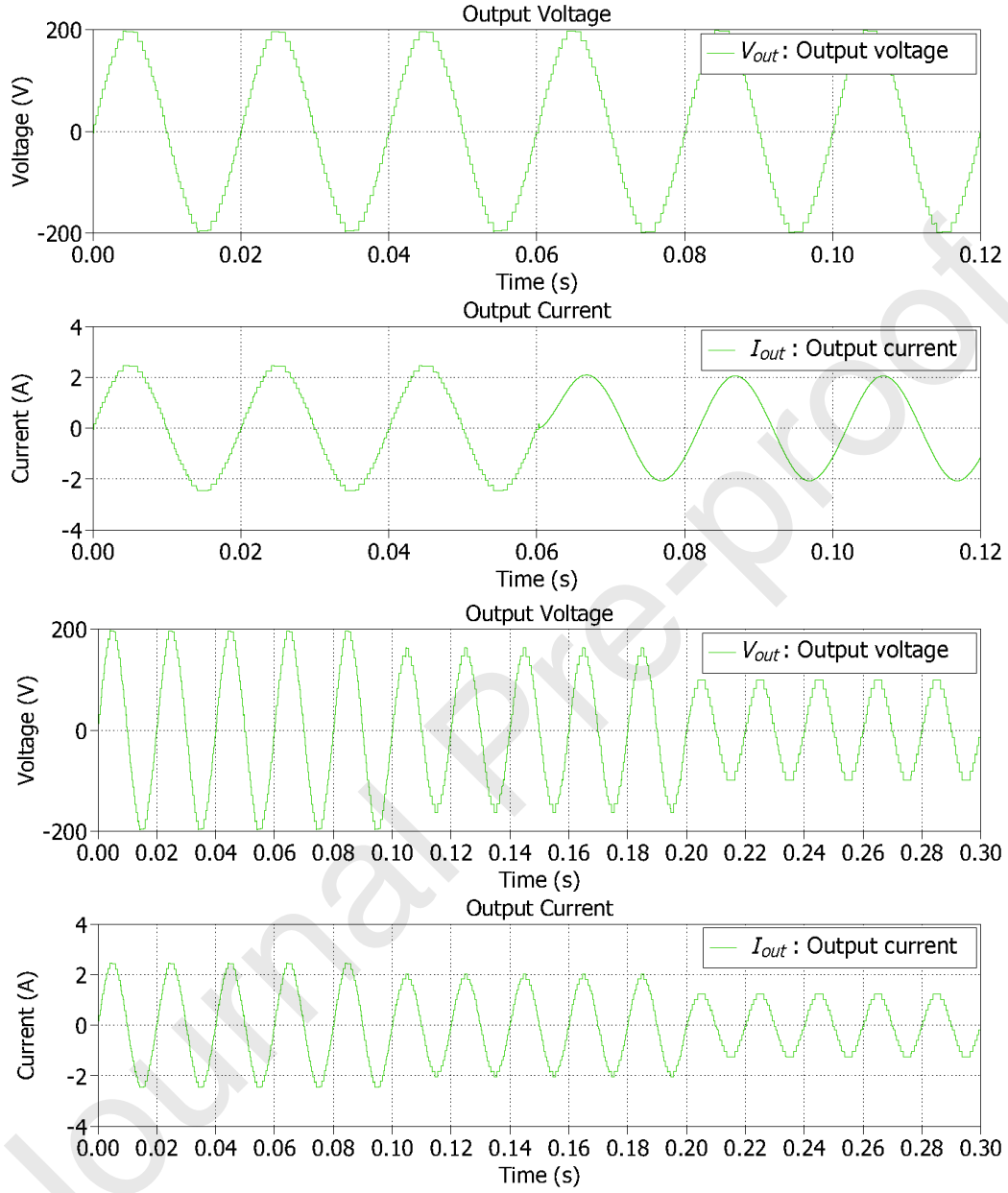


Figure 8: Simulation results of AC output voltage V_{out} and current I_{out} of the proposed SC-MLI with transition in (a) load from resistive load $R = 80\Omega$ to resistive-inductive load $R - L = 80\Omega + 167.5\text{mH}$ and (b) MI from 1.0 to 0.8 to 0.5.

317 a 25-level AC output waveform (V_{out}) with the boost of total input voltage
 318 ($V_{in,total}$) to a maximum AC output voltage ($V_{o,peak}$) of approximately 200V
 319 when driving a resistive-inductive load ($R - L = 80\Omega + 167.5\text{mH}$). Par-
 320 ticularly under resistive-inductive loading, the AC output current waveform
 321 (I_{out}) exhibits a pure sinusoidal shape. The resulting output voltage and
 322 current are measured at $140.05V_{rms}$ and $1.41I_{rms}$, respectively, with THD
 323 values of 3.72% and 0.81%, respectively. Furthermore, it exhibits the sim-
 324 ilar self-balancing behavior of capacitor voltages ($V_{c,d1} = V_{c,u1} = 50V$ and
 325 $V_{c,d2} = V_{c,u2} = 50V$) in accordance with the scenario when driving the pure
 326 resistive load. Additionally, detailed information regarding the blocking volt-
 327 age across different switches in the proposed SC-MLI is provided in Fig. 7,
 328 particularly when the DC strings are set to $V_1 = V_2 = V_3 = 16.6V$ for 25-
 329 level operation. Here, the blocking voltage across switches $S_{L,1}$, $S_{L,2}$, $S_{c,1}$,
 330 $S'_{c,1}$, and bidirectional switches $S_{a,1}$, $S_{a,2}$ is 50V, while switches $S_{c,2}$ and $S'_{c,2}$
 331 exhibit 100V, and switches $S_{R,1}$ and $S'_{R,1}$ experience 200V.

332 The proposed SC-MLI is further tested under dynamic change in load
 333 and MI . Fig. 8(a) illustrates the dynamic load capability of the proposed
 334 topology, transitioning from a purely resistive load ($R = 80\Omega$) to a resistive-
 335 inductive load ($R - L = 80\Omega + 167.5\text{mH}$), without introducing any distortion
 336 in the output voltage and current waveforms. It is noteworthy that during
 337 this transition, the AC current waveform shifts from a 25-level staircase pat-
 338 tern to a pure sinusoidal shape, showcasing the adaptability of the system to
 339 changing load conditions. Furthermore, the proposed topology exhibits the
 340 flexibility to adjust its output characteristics at different modulation index
 341 (MI) ranging from 1.0 to 0.8 to 0.5 under the purely resistive load condition
 342 ($R = 80\Omega$), as depicted in Fig. 8(b). At $MI = 0.8$, the SC-MLI generates a
 343 21-level output waveform with an output voltage of $117.43V_{rms}$ and current
 344 of $1.43I_{rms}$. The THD values for voltage and current are measured at 4.77%
 345 and 4.61%, respectively. Similarly, at $MI = 0.5$, the SC-MLI produces a
 346 13-level output waveform with a voltage output of $70.54V_{rms}$ and current of
 347 $0.87I_{rms}$. At this MI , the THD values for voltage and current are recorded
 348 at 6.94% and 6.77%, respectively. Hence, PLECS simulation studies demon-
 349 strate the feasibility of the proposed topology for renewable or industrial
 350 applications.

6. Experimental Validation

The proposed SC-MLI (25-level) with DC strings configuration is further examined in the experiment to validate its practical applicability. It is tested under two loading conditions, i.e., $R = 80\Omega$ and $R = 80\Omega$ and $L = 167.5mH$ with the control of $MI = \{1.0, 0.8, 0.5\}$. Three DC sources ($V_1 = V_2 = V_3 = 16.6V$) and 2 SC units with capacitors ($C_{d,1} = C_{u,1} = 3300 \mu F$, $C_{d,2} = C_{u,2} = 2200 \mu F$, $C_{c,dc1} = C_{c,dc2} = 1000 \mu F$) are utilized to produce 25-level output voltage and boost the total input voltage from 50V to output peak voltage of 200V. The implementation involves AUIRGP50B60PD1 IGBT switches with anti-parallel diodes, MUR1560G diodes, and NLC modulation in MATLAB/SIMULINK. Gate signals are amplified by HCPL-3120 opto-coupler gate drivers, facilitating IGBT switch operation in the MLI prototype. Note that in this experiment, IGBTs are used instead of MOSFETs to enhance durability and minimize failure risks during testing. Moreover, they offer higher voltage and current handling capability, making them more reliable for experimental research. Additionally, since the inverter operates at a low switching frequency, IGBTs provide better tolerance to surge currents and short-circuit conditions. Capacitance values chosen are also considered to minimize capacitor voltage ripple ($\leq 10\%$) due to its longest discharging cycle. It can be calculated using the formula:

$$C = \frac{I \times \Delta t}{\Delta V_C}. \quad (2)$$

where ΔV_C is the voltage ripple across the capacitor, I is the current through the capacitor during discharging, Δt is the duration of the discharging period, C is the capacitance. Detailed specifications for the proposed topology, including component parameters and equipment types, are presented in Table 5. Fig. 9 illustrates the experimental setup overview for the DC strings-based SC-MLI.

A dynamic load test is conducted to understand the response of the proposed SC-MLI. The load changes from pure resistive to resistive-inductive loads with the control of MI . The fundamental frequency used in the experiment is $50Hz$. Fig. 10 depicts the AC output voltage (V_{out}) and current (I_{out}) waveforms, including the harmonic profiles under pure resistive $R = 80\Omega$ at 500W. These figures reveal that the proposed SC-MLI effectively converts DC inputs into 25-level AC output waveforms (V_{out} and I_{out}), thereby boosting the total input voltage ($V_{in,total}$) from 50V to

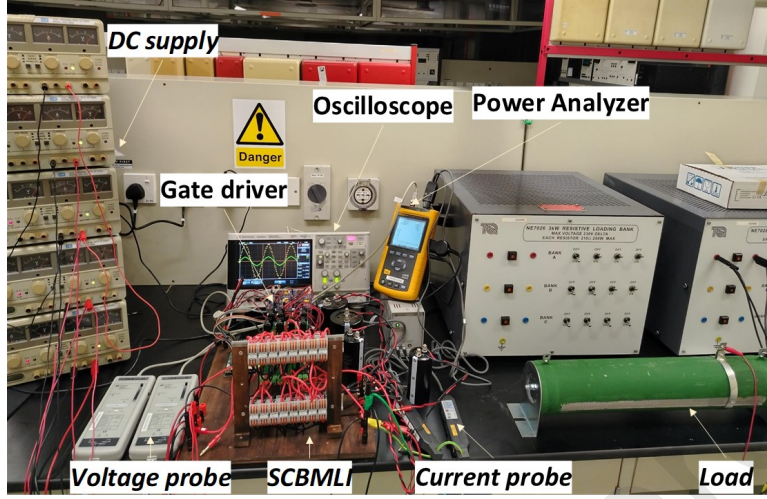


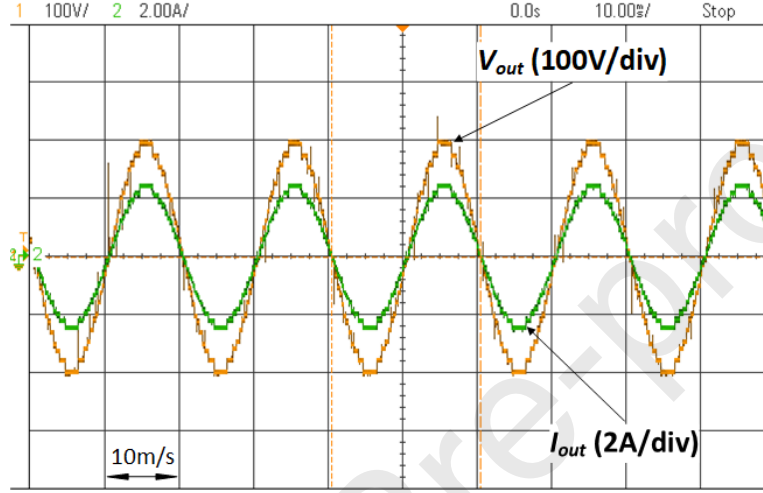
Figure 9: Lab-scale experimental setup for the DC strings based SC-MLI performance verification.

Table 5: Specification of the Proposed Topology (25-level) in Measurement.

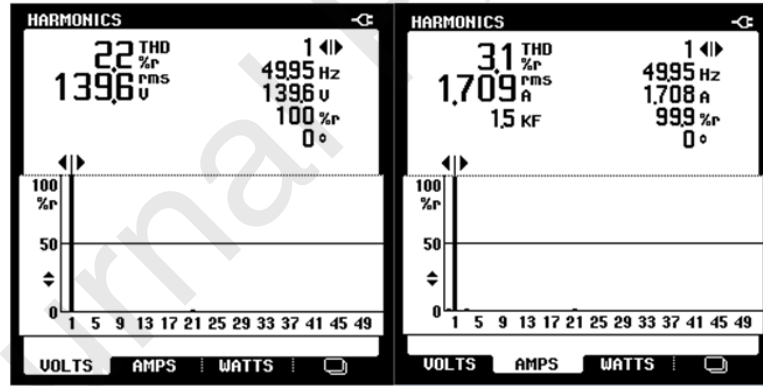
Description	Value/Parameters
Input DC source	$V_1 = V_2 = V_3 = 16.6V$
Boost gain	$V_b = 4$
Fundamental frequency	$f = 50 \text{ Hz}$
Loads	$R \text{ load: } 80\Omega$
	$R - L \text{ load: } 80\Omega \text{ and } 167.5\text{mH}$
Capacitor	$C_{d,1} = C_{u,1} = 3300 \mu\text{F}, C_{d,2} = C_{u,2} = 2200 \mu\text{F}$
Switches (IGBT)	AUIRGP50B60PD1
Diodes	MUR1560G
Optocoupler Gate Driver	HCPL-3120
DSP Controller	EZDSP F28335
Voltage probe	Keysight N2891A
Current probe	Keysight N2783B AC/DC
Oscilloscope	DSO-X 2012A
Power Analyzer	Fluke 43B
DC Power Supply	GW Instek GPC-3030B

385 a maximum AC output voltage ($V_{o,peak}$) of approximately 200V. The out-
 386 put voltage and current are $139.8V_{rms}$ and $1.709I_{rms}$ with a conversion ef-

387 efficiency of 95.43%. The THD values for both the 25-level output voltage
 388 and current are 2.2% and 3.1%, respectively. Furthermore, Fig. 11 illus-
 389 trates the self-balancing of capacitor voltages around their respective values
 390 ($V_{c,d1} = V_{c,u1} = 50V, V_{c,d2} = V_{c,u2} = 50V$) without the need for auxiliary
 391 circuits or control systems.



(a)



(b)

(c)

Figure 10: Experimental results of (a) AC output voltage V_{out} and current I_{out} , (b) harmonic profile of AC output and (c) current of the proposed 25-level SC-MLI under resistive load $R = 80\Omega$.

392 Similarly, the proposed SC-MLI is capable of generating a 25-level AC
 393 output waveform (V_{out}) and boosting the total input voltage ($V_{in,total}$) to a

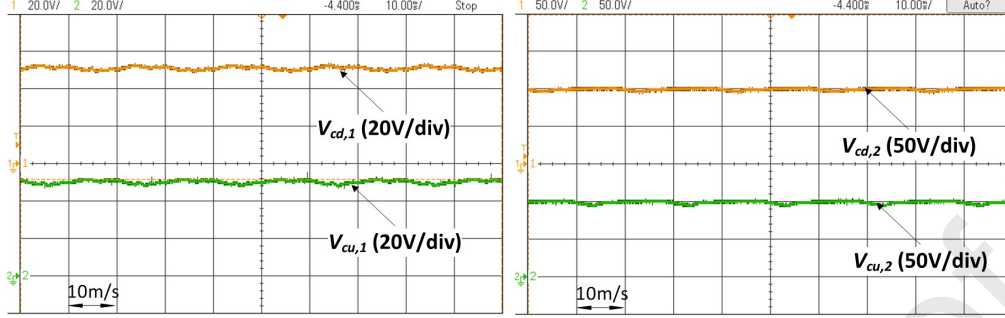


Figure 11: Experimental results of capacitor voltage $V_{c,d1}$, $V_{c,u1}$, $V_{c,d2}$, and $V_{c,u2}$ in the proposed SC-MLI under resistive load $R = 80\Omega$.

maximum AC output voltage ($V_{o,peak}$) of around 200V when driving resistive-inductive loading ($R - L = 80\Omega + 167.5\text{mH}$) as shown in Fig. 12. Notably, under $R - L$ loading, the AC output current waveform (I_{out}) exhibits a pure sinusoidal shape. The resulting output voltage and current are $139.7V_{rms}$ and $1.343I_{rms}$, with THD values of 1.5% and 1.4%, respectively. Furthermore, Fig. 13 illustrates the similar self-balancing of capacitor voltages ($V_{c,d1} = V_{c,u1} = 50V$, $V_{c,d2} = V_{c,u2} = 50V$), closely matching the scenario depicted in Fig. 11. Additionally, details regarding the blocking voltage across different switches in the proposed SC-MLI are available in Fig. 14 when the DC strings are set to $V_1 = V_2 = V_3 = 16.6V$ for 25-level operation. The blocking voltage across switches $S_{L,1}$, $S_{L,2}$, $S_{c,1}$, $S'_{c,1}$, and bidirectional switches $S_{a,1}$, $S_{a,2}$ is 50V; switches $S_{c,2}$ and $S'_{c,2}$ exhibit 100V; switches $S_{R,1}$ and $S'_{R,1}$ experience 200V. Table 6 presents the measured switching frequencies of each switch in the proposed SC-MLI. The left-hand bridge switches, responsible for multiple output voltage level creation, operate at switching frequencies of 100 Hz for $S_{L,1}$, 86 Hz for $S_{L,2}$, 769.2 Hz for $S_{a,1}$ and $S'_{a,1}$, and 1030 Hz for $S_{a,2}$ and $S'_{a,2}$. The SC switches, responsible for charging and discharging capacitors to boost the voltage, operate at 507.4 Hz for $S_{c,1}$ and $S'_{c,1}$, and 263.7 Hz for $S_{c,2}$ and $S'_{c,2}$. The right-hand bridge switches, responsible for DC-to-AC conversion, exhibit switching frequencies of 50 Hz for $S_{R,1}$ and $S'_{R,1}$.

Fig. 15(a) depicts the capability of the proposed topology to perform dynamic load change from $R = 80\Omega$ to $R - L = 80\Omega + 167.5\text{mH}$ without any distortion in output voltage and current waveforms. Notably, during this transition, the AC current shifts from its 25-level staircase waveform to a pure sinusoidal shape, highlighting the adaptability of the system to varying load conditions. Moreover, the proposed topology demonstrates the ability to

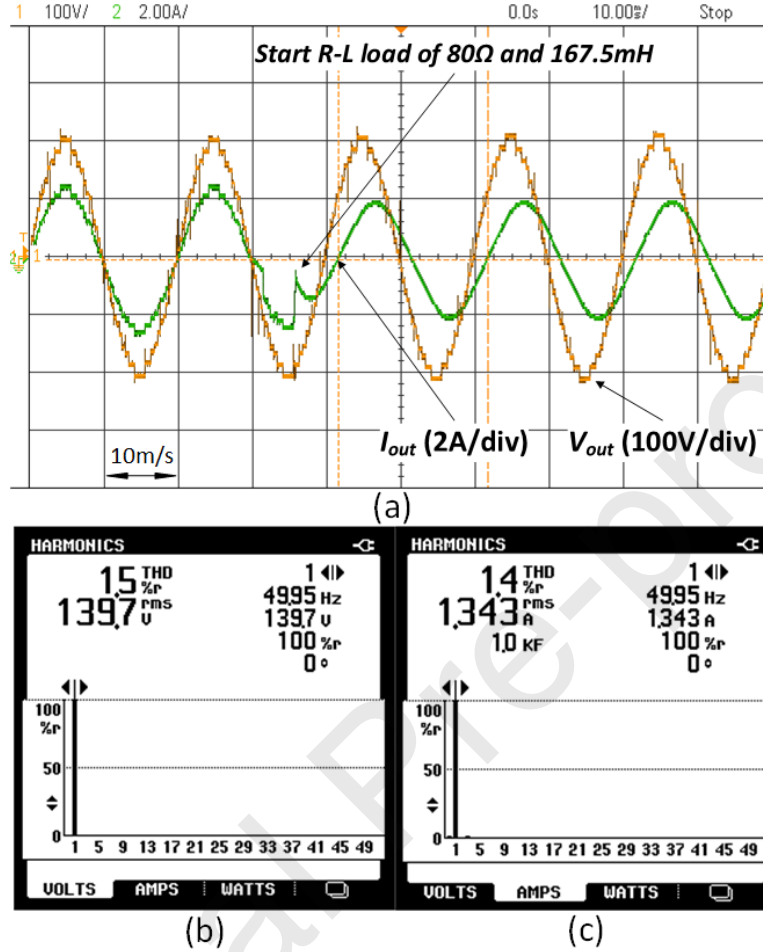


Figure 12: Experimental results of (a) AC output voltage V_{out} and current I_{out} , (b) harmonic profile of AC output and (c) current of the proposed 25-level SC-MLI under resistive-inductive load $R - L = 80\Omega + 167.5\text{mH}$.

adjust its output amplitude and level to desired values at different MI from 1.0 to 0.8 to 0.5 under $R = 80\Omega$, as depicted in Fig. 15(b). At $MI = 0.8$, the proposed SC-MLI generates a 21-level output waveform with output voltage and current measuring $116.5V_{rms}$ and $1.433I_{rms}$, respectively. The associated THD values for voltage and current stand at 2.3% and 3.1%, respectively. Similarly, at $MI = 0.5$, the SC-MLI generates a 13-level output waveform with an output voltage of $70.2V_{rms}$ and a current of $0.858I_{rms}$. The THD values for voltage and current at this MI are 5.5% and 7.5%, respectively.

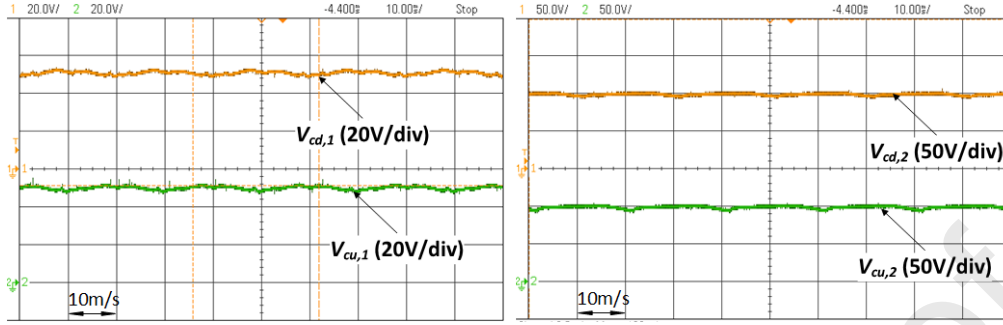


Figure 13: Experimental results of capacitor voltage $V_{c,d1}$, $V_{c,u1}$, $V_{c,d2}$, and $V_{c,u2}$ in the proposed SC-MLI under resistive-inductive load $R - L = 80\Omega + 167.5\text{mH}$.

Table 6: Measured Switching Frequencies of Different Switch Types

Switch Type	Switches	Measured Switching Frequency (Hz)
Left-hand bridge switches	$S_{L,1}$	100.0
	$S_{a,1}, S_{a,1'}$	769.2
	$S_{a,2}, S_{a,2'}$	1030.0
	$S_{L,2}$	86.0
Switched capacitor switches	$S_{c,1}, S_{c,1'}$	507.4
	$S_{c,2}, S_{c,2'}$	263.7
Right-hand bridge switches	$S_{R,1}, S_{R,1'}$	50.0

Throughout the experiment, it is proven that the proposed SC-MLI can respond to dynamic changes in modulation index and load conditions without introducing distortion. This makes the inverter suitable for AC output applications with closed-loop control, as the voltage amplitude can be effectively controlled by adjusting the modulation index. The ability to precisely regulate the output of an inverter ensures its adaptability to various control systems, including future applications like STATCOM systems, where dynamic voltage control and load compensation are crucial. For instance, for a STATCOM connected at the point of common coupling of the grid, when the grid voltage is measured at 240V, no current needs to flow through the coupling inductor/transformer. In other words, the inverter will produce 240V at a modulation index of 0.5 to match the grid voltage, preventing any current flow through the coupling inductor/transformer. Thus, the inverter does not need to generate or absorb current through its coupling inductor/transformer. However, when the grid voltage sags due to a high load

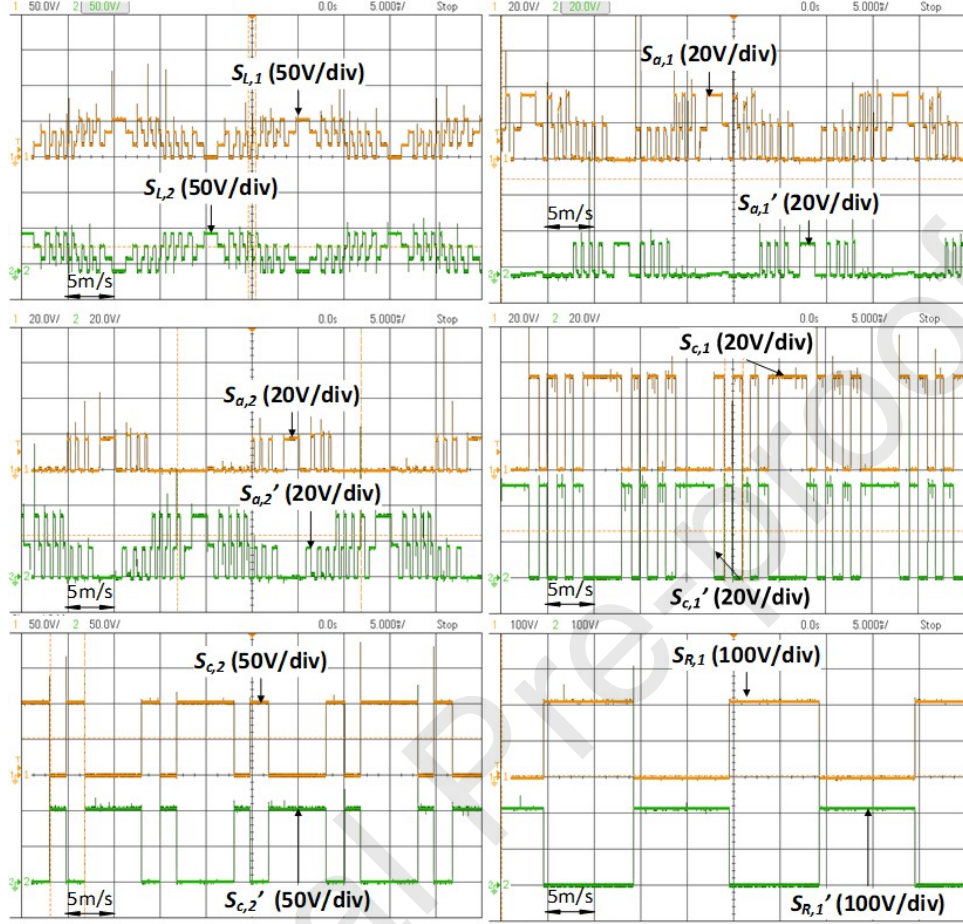


Figure 14: Experimental results of the voltage stress for the proposed SC-MLI in 25-level DC strings configuration.

power factor, the inverter must generate higher voltage amplitudes with a modulation index in the range of $0.5 < MI \leq 1$ to inject reactive current into the grid via the coupling inductor/transformer for reactive power compensation. Conversely, during a grid voltage swell caused by a low load power factor, the inverter needs to generate lower voltage amplitudes with a modulation index in the range of $0 \leq MI < 0.5$ to draw/absorb reactive current from the grid [36]. Hence, the proposed extendable SC-MLI can be configured to generate at least 17 voltage levels under these conditions, complying with IEEE standards with a THD percentage of less than 5%. It is important to clarify that while the discussion above illustrates the potential role of

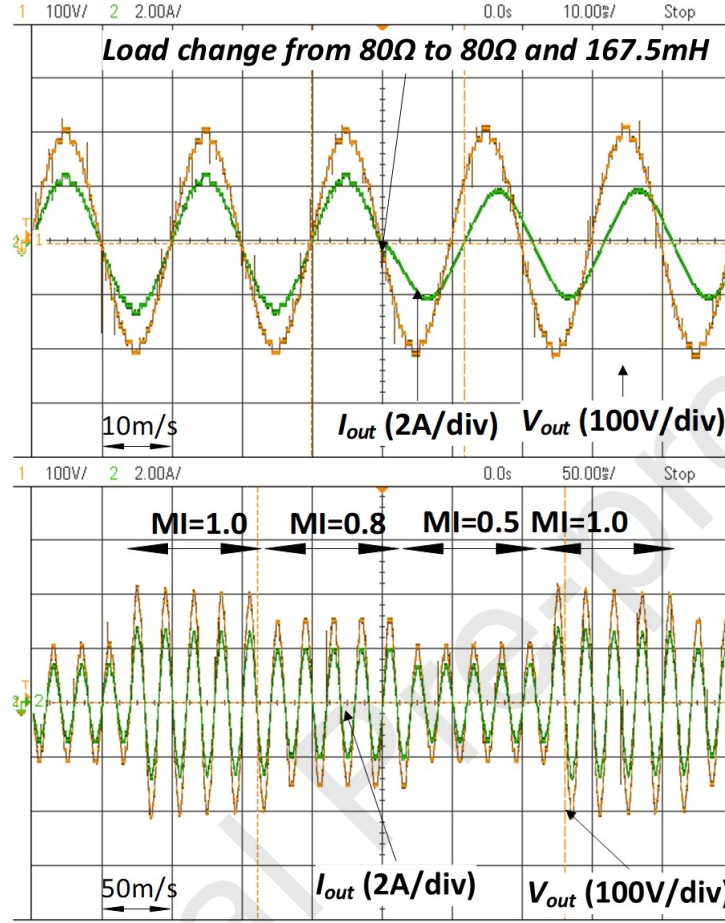


Figure 15: Experimental results of AC output voltage V_{out} and current I_{out} of the proposed SC-MLI with transition in (a) load from resistive load $R = 80\Omega$ to resistive-inductive load $R - L = 80\Omega + 167.5mH$ and (b) MI from 1.0 to 0.8 to 0.5.

the proposed SC-MLI in STATCOM applications, this paper does not implement or simulate a complete STATCOM control system. The experimental and simulation results presented here are limited to the inverter operating independently under varying load and modulation index conditions. Modeling, control integration, and performance evaluation of the SC-MLI within a STATCOM setup are beyond the current scope and are suggested as avenues for future research.

460 6.1. Power Loss Analysis

461 The power loss and efficiency of the proposed SC-MLI are analyzed using
 462 PLECS software. The simulation uses real data from the device manufac-
 463 turer for all power devices, including switches and diodes, and incorporates
 464 the thermal model for the proposed topologies. The simulation is conducted
 465 under steady-state conditions, with an ambient temperature of 25°C across
 466 the heat sink. It is worth noting that the switches and diodes used in this
 467 study are C3M0025065K (650V/97A) and C3D16060D (600V/22A), respec-
 468 tively.

469 In power analysis of SC-MLIs, the total power loss P_{TL} and efficiency η
 470 are determined by considering switching P_{sw} and conduction P_c losses from
 471 the switches and diodes, along with the capacitor ripple P_{RP} , and resistive
 472 losses P_r from switched-capacitor units can be calculated by the following
 473 equation [26, 28, 27, 29, 30, 31, 32].

$$P_{TL} = P_{sw} + P_c + P_{RP} + P_r. \quad (3)$$

474 6.1.1. Switching losses (P_{sw})

475 The switching losses occur when the voltage and current through the
 476 switches are overlapping each other due to non-ideal behavior of the switches
 477 during the ON and OFF state.

$$P_{sw,on_i} = \frac{1}{6}f \times V_{sw,on_i} \times I_{sw,on_i} \times t_{on}, \quad (4)$$

$$P_{sw,off_i} = \frac{1}{6}f \times V_{sw,off_i} \times I_{sw,off_i} \times t_{off}, \quad (5)$$

478 where f is the fundamental frequency, t_{on} and t_{off} are the time taken by
 479 the switches voltage, current, and power to turn ON ($V_{sw,on,i}$, $I_{sw,on,i}$, and
 480 $P_{sw,on,i}$), and OFF ($V_{sw,off,i}$, $I_{sw,off,i}$, and $P_{sw,off,i}$) during switching, respec-
 481 tively. Therefore, the switching losses of i -th number of switches from the
 482 SC-MLI can be computed by multiplying the total number of ON (N_{on}) and
 483 OFF (N_{off}) of the switching states as follows

$$P_{sw,total} = \sum_{i=1}^{N_{sw}} \left(\sum_{j=1}^{N_{on}} P_{sw,on_{ij}} + \sum_{j=1}^{N_{off}} P_{sw,off_{ij}} \right). \quad (6)$$

484 *6.1.2. Conduction losses (P_c)*

485 There are conduction losses from switches and diodes with turn ON-state
 486 internal resistance. Thus, the total conduction losses of these power devices
 487 can be calculated by

$$P_{c,total} = \sum_{(\forall SW)} I_{avg,switch}^2 R_{on} + \sum_{\forall D} I_{avg,diode}^2 R_d, \quad (7)$$

488 where R_{on} and R_d are the internal turn ON-state resistance of the switch
 489 and diode, respectively, $I_{avg,switch}$ and $I_{avg,diode}$ are the average current flow
 490 via the switch and diode during the ON-state, respectively.

491 *6.1.3. Capacitor ripple and resistive losses (P_{RP} and P_r)*

492 The capacitor ripple and resistive losses (P_{RP} and P_r) occur due to the
 493 voltage drop ΔV_c and power dissipation from the internal resistance R_c of
 494 the capacitors during the capacitors charging and discharging. The total
 495 capacitor ripple losses can be expressed as follows

$$P_{RP,total} = \sum_{(\forall Caps)} \Delta V_c^2 \times \pi \times f_{(c,sw)} \times C \tan \delta. \quad (8)$$

496 The total capacitor resistive losses can be calculated by,

$$P_r = \sum_{(\forall Caps)} I_{charging}^2 R_c + I_{discharging}^2 R_c, \quad (9)$$

497 where $f_{c,sw}$ is capacitor switching frequency and δ is dissipation factor which
 498 is set to 0.2. The overall system efficiency of the proposed SC-MLIs can be
 499 calculated as follows

$$\eta = \frac{P_{in} - (P_{sw} + P_c + P_{RP} + P_r)}{P_{in}}. \quad (10)$$

500 where P_{in} and P_{out} refers to the input source and output power.

501 The power loss distribution of the proposed DC-strings-based SC-MLI in
 502 25 levels at 500W in Table 7 is illustrated for each component. It can be
 503 observed that the majority of power losses from the proposed topology origi-
 504 nate from switched-capacitor units. Notably, the power losses of switches
 505 and diodes allocated at the switched-capacitor units of the proposed topol-
 506 ogy exhibit higher power losses compared to other power devices due to the

capacitor charging flow. However, the proposed topology is able to exhibit high efficiency of 97.31% with only a total power loss of 13.53W.

When comparing the proposed topology with recently developed extendable SC-MLIs [26, 28, 27, 29, 30, 31, 32] at a power load of 100W to 1000W, as outlined in Table 8. It is evident that the proposed SC-MLI demonstrates higher efficiency among all the SC-MLIs. It can be attributed to the placement of capacitors on the upper and bottom sides, allowing them to charge and discharge alternately throughout the full cycle. Consequently, capacitor voltage drop and charging current are minimized, leading to reduced total capacitor losses. Moreover, the DC strings design eliminates the need for power devices connected between the DC sources, thereby reducing capacitor power losses from the charging current across the switches. The simulation and experimental efficiency of the proposed topology at 100W to 500W are shown in Fig. 16. It is observed that the proposed topology has the highest efficiency of 98.68% for simulations and 96.46% for experiments.

Table 7: Power loss distribution of the proposed SC-MLI at 500W.

Switches	P_{TL} (W)	Loss (%)	Diodes	P_{TL} (W)	Loss (%)
$S_{L,1}$	0.02	0.14	$D_{c,1}$	2.54	18.76
$S_{a,1}$	0.04	0.27	$D'_{c,1}$	2.55	18.84
$S_{a,2}$	0.04	0.27	$D_{c,2}$	0.91	6.75
$S_{L,2}$	0.02	0.14	$D'_{c,2}$	0.92	6.81
$S_{c,1}$	0.42	3.12	Capacitors	P_{TL} (W)	Loss %
$S'_{c,1}$	0.41	3.05			
$S_{c,2}$	0.17	1.25			
$S'_{c,2}$	0.16	1.20			
$S_{R,1}$	0.04	0.27			
$S'_{R,1}$	0.04	0.28			
L_T (w)	13.53				
η (%)	97.31				

7. Conclusion

A novel DC strings-based SC-MLI has been proposed to offer the flexibility to extend with basic DC sources and SC units. The multi-source config-

Table 8: Comparative study of SC-MLIs in terms of the calculated efficiency.

Topology	Ref.	N_L	V_B	P(W)	η (%)
Ours		25	4	100	98.68
				200	98.32
				300	97.98
				400	97.65
				500	97.31
				1000	95.64
Barzegarkhoo et al., 2016	[26]	17	2	1000	90
Hassan et al., 2020	[27]	21	1.71	100~500	96.5~94.5
Ramaiah et al., 2020	[28]	25	4	454	90.8
Bana et al., 2021	[29]	25	2	100~400	96.0~94.5
Hosseinzadeh et al., 2022	[30]	25	4	500	78.5
Singh et al., 2023	[31]	19	1.57	560	96.5
Singh et al., 2023	[32]	21	3.33	1000	91.15

525 uration can provide an output voltage level creation during the initial phase,
 526 while the series-parallel combination of switched capacitors efficiently boosts
 527 up the input voltage without requiring an auxiliary circuit. Our proposed
 528 topology uses only 12 switches in achieving a 25 voltage levels and four times
 529 boosting capability. It has an efficiency of more than 97.31% at 500W as
 530 compared to the other developed extendable SC-MLIs. The proposed topol-
 531 ogy has been tested in both simulation and experiment with dynamic load
 532 tests from resistive to resistive-inductive load with the varying modulation
 533 index. Moreover, the proposed SC-MLI complies with the IEEE standards
 534 with a THD percentage of less than 5%. These findings not only demon-
 535 strate the effectiveness of the methodology but also emphasize its potential
 536 for practical implementation in real-world applications such as STATCOM
 537 within the field of power electronics.

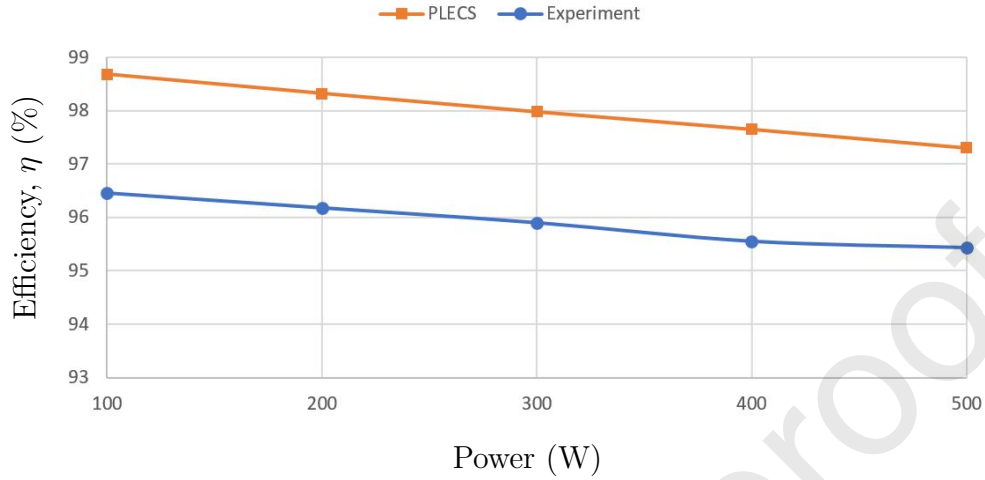


Figure 16: PLECS and experiment efficiency of the proposed SC-MLI in 25-level DC strings configuration.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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